Topic

Indexing Structures for Files

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Chapter Outline

- Types of Single-level Ordered Indexes
  - Primary Indexes
  - Clustering Indexes
  - Secondary Indexes
- Multilevel Indexes
- Dynamic Multilevel Indexes Using B-Trees and B+-Trees
- Indexes on Multiple Keys

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Indexes as Access Paths

- A single-level index is an auxiliary file that makes it more efficient to search for a record in the data file.
- The index is usually specified on one field of the file (although it could be specified on several fields)
- One form of an index is a file of entries **<field value, pointer to record>,** which is ordered by field value
- The index is called an *access path* on the field.

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Indexes as Access Paths (contd.)

- The index file usually occupies considerably less disk blocks than the data file because its entries are much smaller
- A binary search on the index yields a pointer to the file record
- Indexes can also be characterized as dense or sparse.
  - A **dense index** has an index entry for *every search key value* (and hence every record) in the data file.
  - A **sparse** (or **nondense**) **index**, on the other hand, has index entries for only some of the search values

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Types of Single-Level Indexes

- **Primary Index**
  - Defined on an ordered data file
  - The data file is ordered on a *key field*
  - Includes one index entry *for each block* in the data file; the index entry has the key field value for the *first record* in the block, which is called the *block anchor*
  - A similar scheme can use the *last record* in a block.
  - A primary index is a nondense (sparse) index, since it includes an entry for each disk block of the data file and the keys of its anchor record rather than for every search value.

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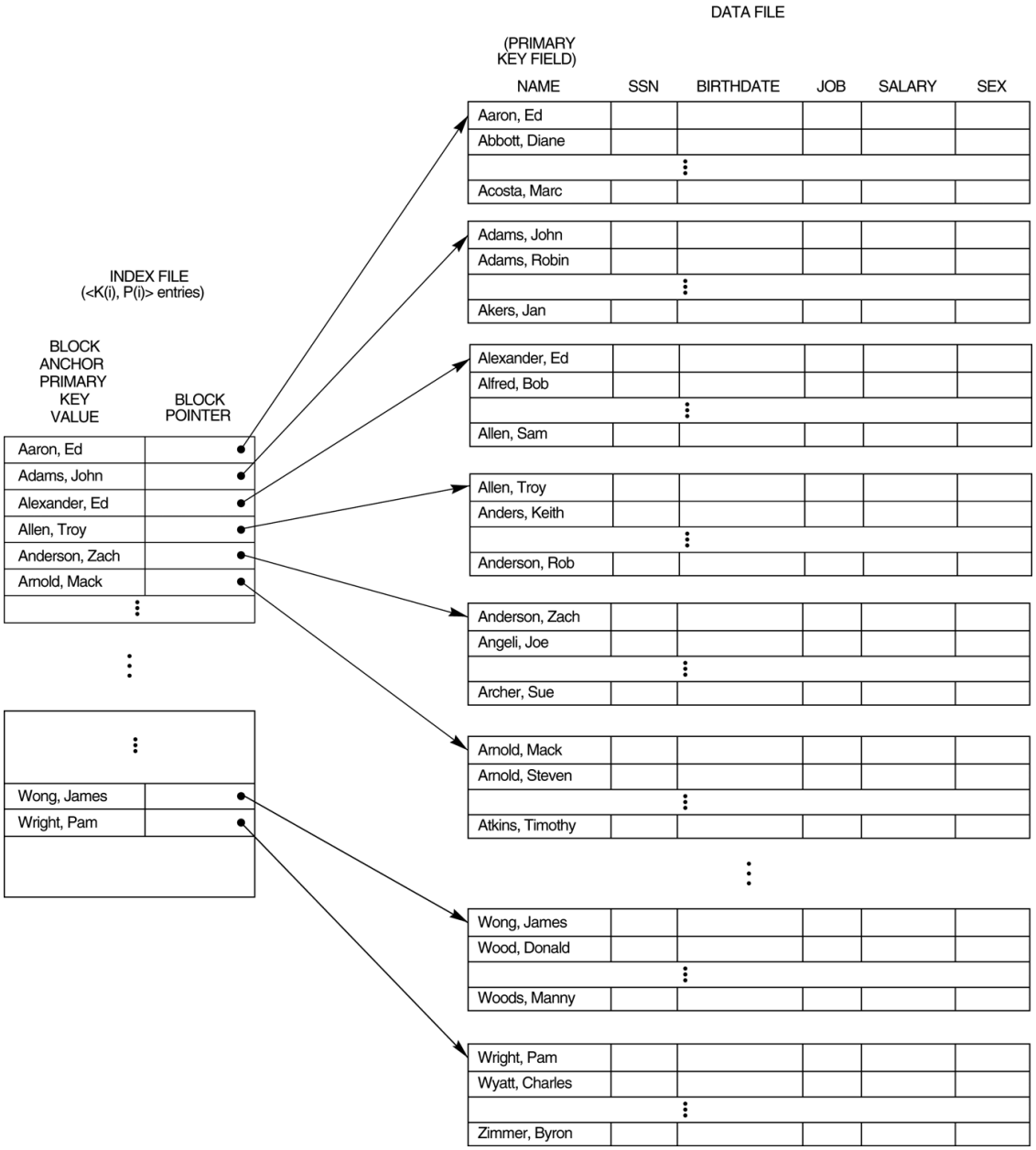
FIGURE 14.1
Primary index on the ordering key field of the file shown in Figure 13.7.



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Primary Index

Example: Given the following data file:
EMPLOYEE(NAME, SSN, ADDRESS, JOB, SAL, ...)
Suppose that:
record size R=100 bytes
block size B=1024 bytes
r=30000 records

Then, we get:
blocking factor bfr= B / R= 1024 / 100= 10 records/block
number of file blocks b= (r/bfr)= (30000/10)= 3000 blocks

For an index on the SSN key field, assume the field size V_{SSN}=9 bytes,
assume the record pointer size P_R=6 bytes. Then:
index entry size $R_I=(V_{SSN}+ P_R)=(9+6)=15$ bytes
index blocking factor $bfr_I= B_I / R_I= 1024 / 15= 68$ entries/block
number of index blocks $b_I= (b/ bfr_I)= (3000/68)= 45$ blocks
binary search needs $\log_2 b_I= \log_2 45= 6$ block accesses, total 6+1=7 block accesses
This is compared to the binary search cost of:

$\log_2 b= \log_2 3000= 12$ block accesses

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Types of Single-Level Indexes

- **Clustering Index**
  - Defined on an ordered data file
  - The data file is ordered on a *non-key field* unlike primary index, which requires that the ordering field of the data file have a distinct value for each record.
  - Includes one index entry *for each distinct value* of the field; the index entry points to the first data block that contains records with that field value.
  - It is another example of *nondense* index where Insertion and Deletion is relatively straightforward with a clustering index.

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FIGURE 14.2

A clustering index on the DEPTNUMBER ordering nonkey field of an EMPLOYEE file.

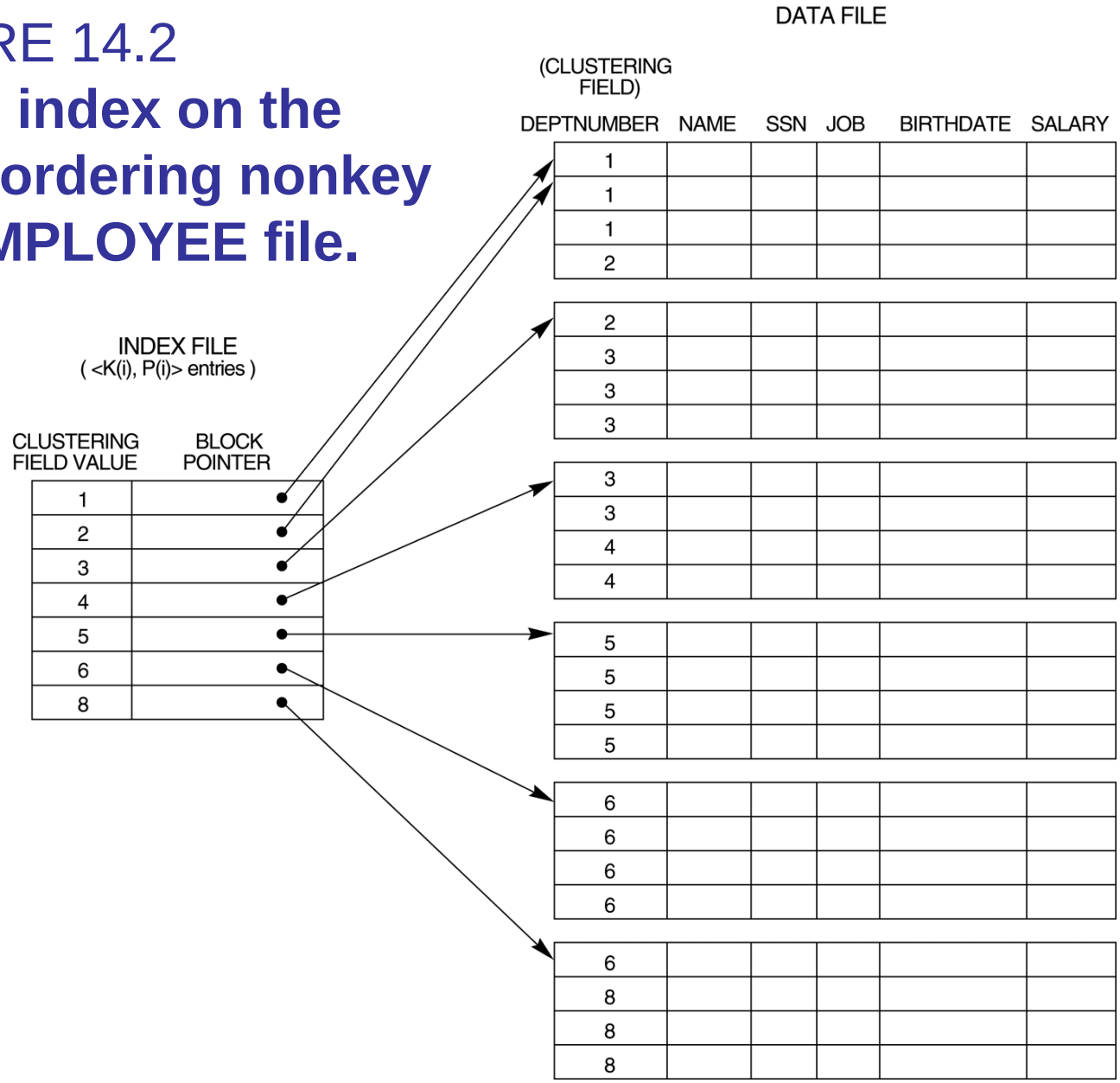

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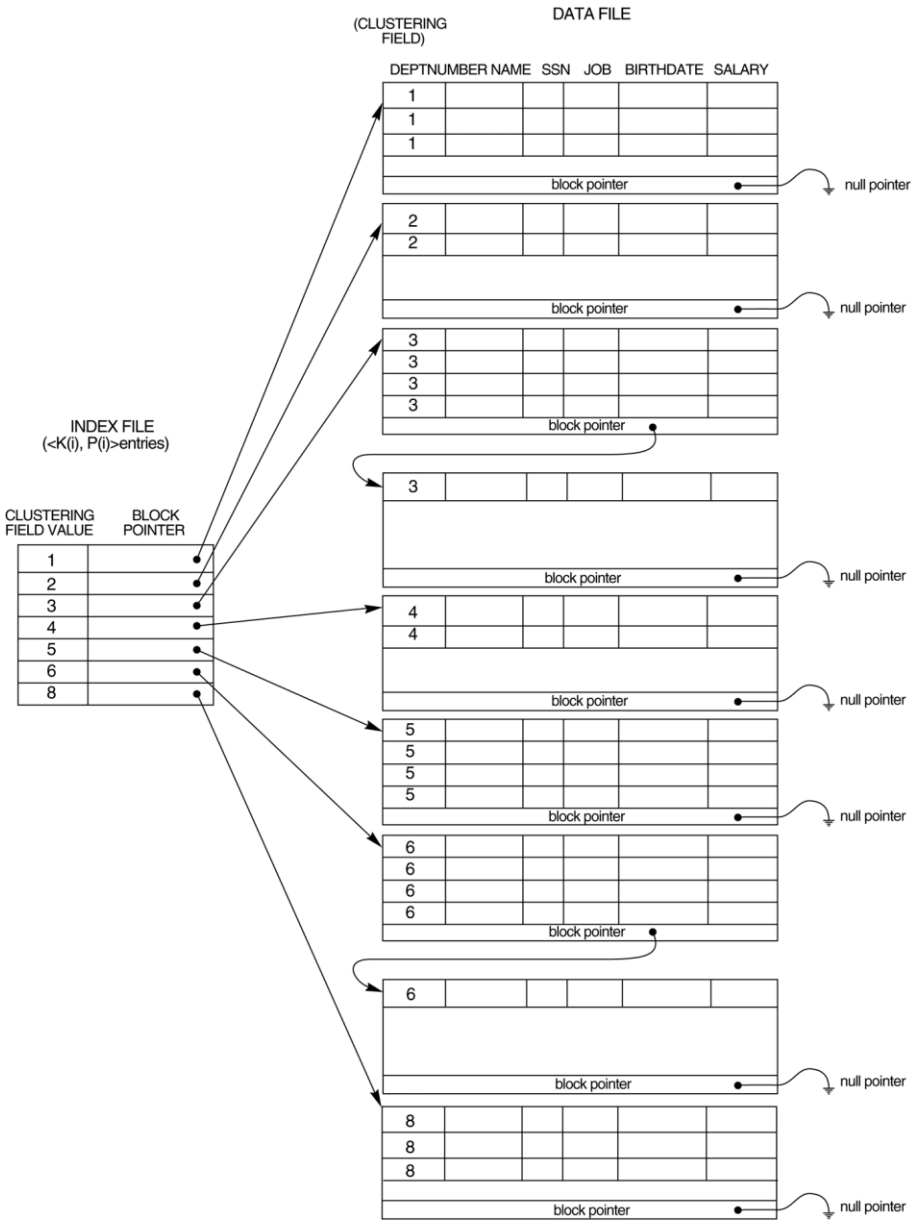
FIGURE 14.3

Clustering index with a separate block cluster for each group of records that share the same value for the clustering field.



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Types of Single-Level Indexes

- **Secondary Index**
  - A secondary index provides a secondary means of accessing a file for which some primary access already exists.
  - The secondary index may be on a field which is a candidate key and has a unique value in every record, or a nonkey with duplicate values.
  - The index is an ordered file with two fields.
    - The first field is of the same data type as some *nonordering field* of the data file that is an *indexing field*.
    - The second field is either a *block* pointer or a *record* pointer. There can be *many* secondary indexes (and hence, indexing fields) for the same file.
  - Includes one entry *for each record* in the data file; hence, it is a *dense index*

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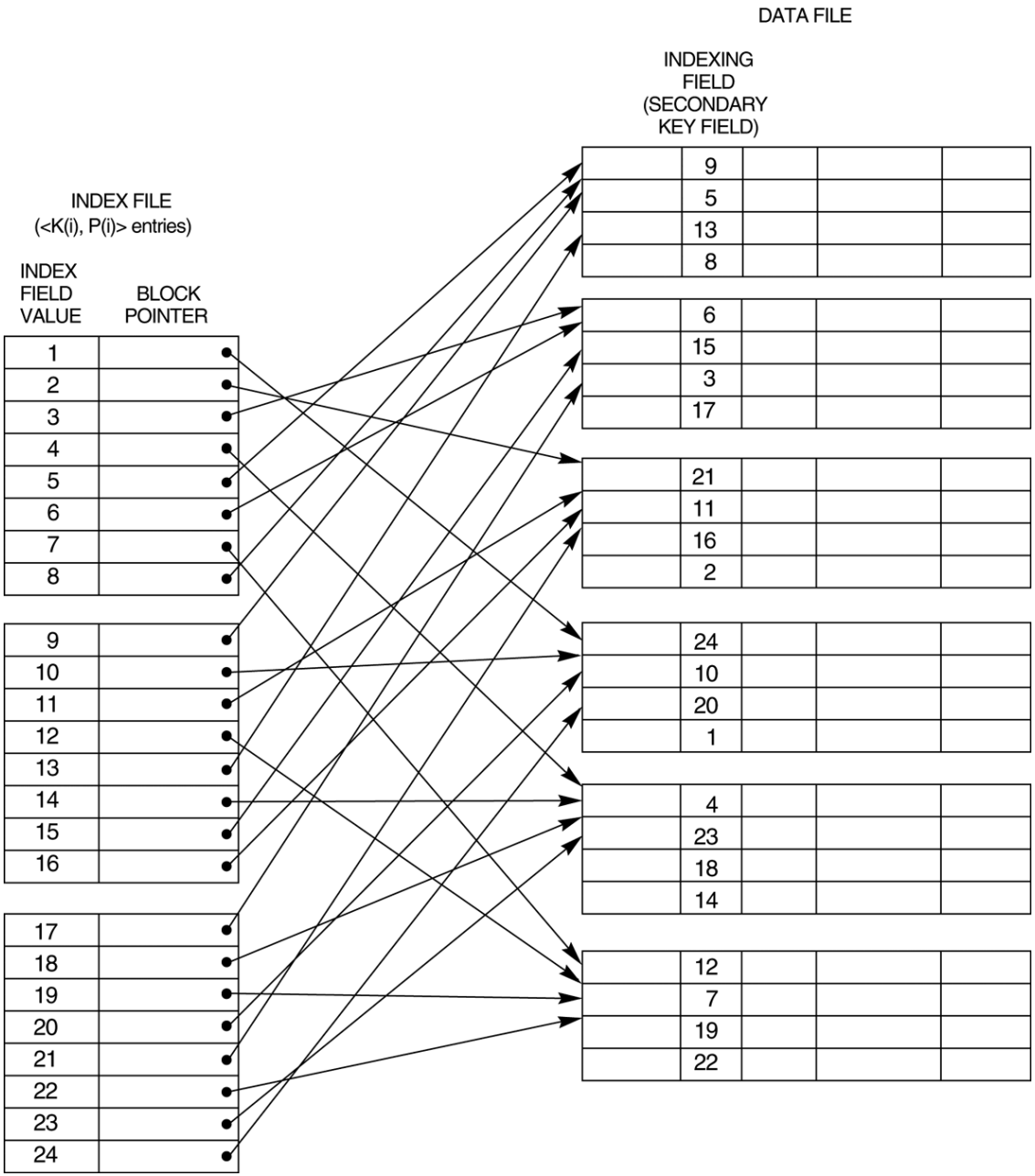
FIGURE 14.4

A dense secondary index (with block pointers) on a nonordering key field of a file.



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Secondary Index

Example: Given the following data file:
EMPLOYEE(NAME, SSN, ADDRESS, JOB, SAL, ...)
Suppose that:
record size R=100 bytes
block size B=1024 bytes
r=30000 records

Then, we get:
blocking factor Bfr= B div R= 1024 div 100 = 10 records/block
number of file blocks b= (r/Bfr)= (30000/10)= 3000 blocks

For an index on the JOB field, assume the field size V_{JOB}=9 bytes,
assume the record pointer size P_R=6 bytes. Then:
index entry size $R_I=(V_{JOB}+ P_R)=(9+6)=15$ bytes
index blocking factor $Bfr_I= B_I$ div R_I= 1024 div 15= 68 entries/block
number of index blocks $b_I= (r/ Bfr_I)= (30000/68)= 442$ blocks
binary search needs $\log_2 b_I= \log_2 442= 9$ block accesses, total 9+1=10 block accesses

This is compared to an average linear search cost of:
(b/2)= 3000/2= 1500 block accesses

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FIGURE 14.5

A secondary index (with recored pointers) on a nonkey field implemented using one level of indirection so that index entries are of fixed length and have unique field values.

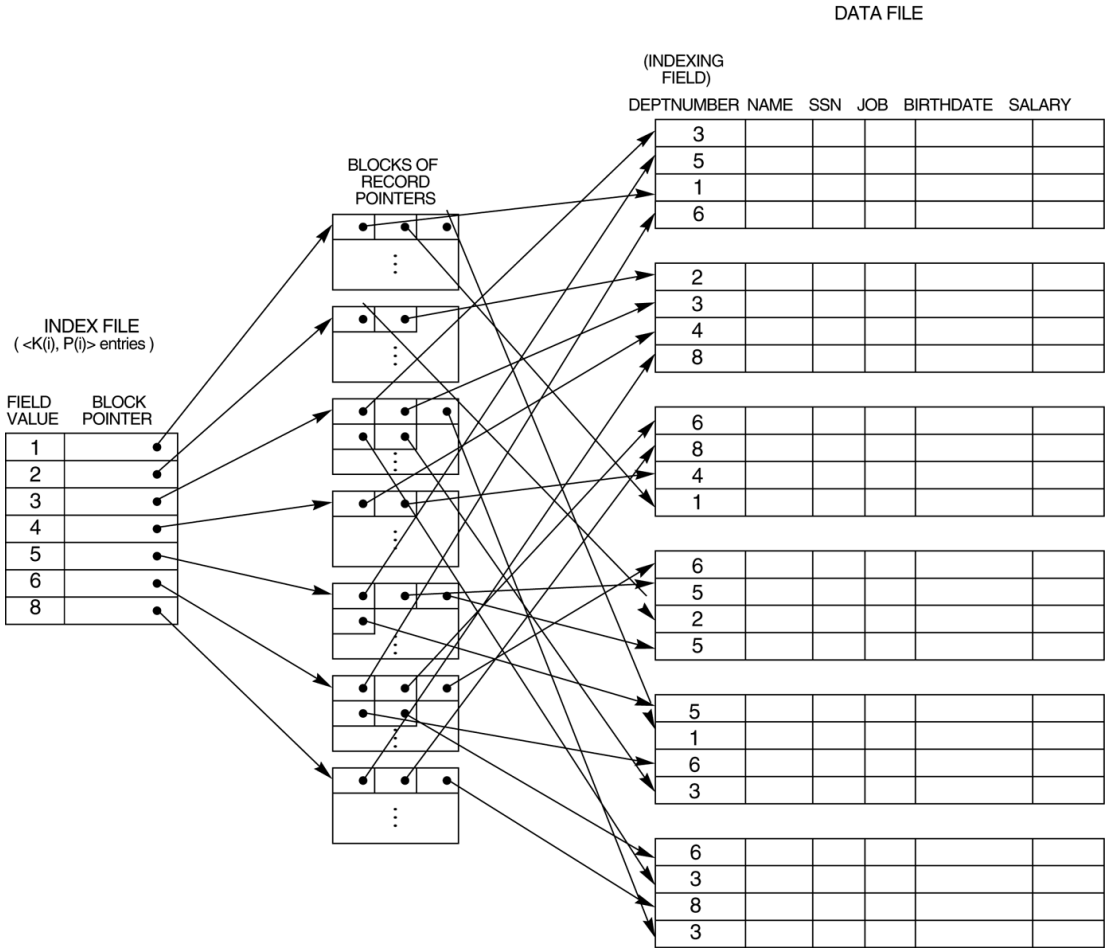

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TABLE 14.2 PROPERTIES OF INDEX TYPES

TYPE OF INDEX	NUMBER OF (FIRST-LEVEL) INDEX ENTRIES	DENSE OR NONDENSE	BLOCK ANCHORING ON THE DATA FILE
Primary	Number of blocks in data file	Nondense	Yes
Clustering	Number of distinct index field values	Nondense	Yes/no[a]
Secondary (key)	Number of records in data file	Dense	No
Secondary (nonkey)	Number of records[b] or Number of distinct index field values[c]	Dense or Nondense	No

[a]Yes if every distinct value of the ordering field starts a new block; no otherwise.
[b]For option 1.
[c]For options 2 and 3.

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Multi-Level Indexes

- Because a single-level index is an ordered file, we can create a primary index *to the index itself* ; in this case, the original index file is called the *first-level index* and the index to the index is called the *second-level index*.
- We can repeat the process, creating a third, fourth, ..., top level until all entries of the *top level* fit in one disk block
- A multi-level index can be created for any type of first-level index (primary, secondary, clustering) as long as the first-level index consists of *more than one* disk block

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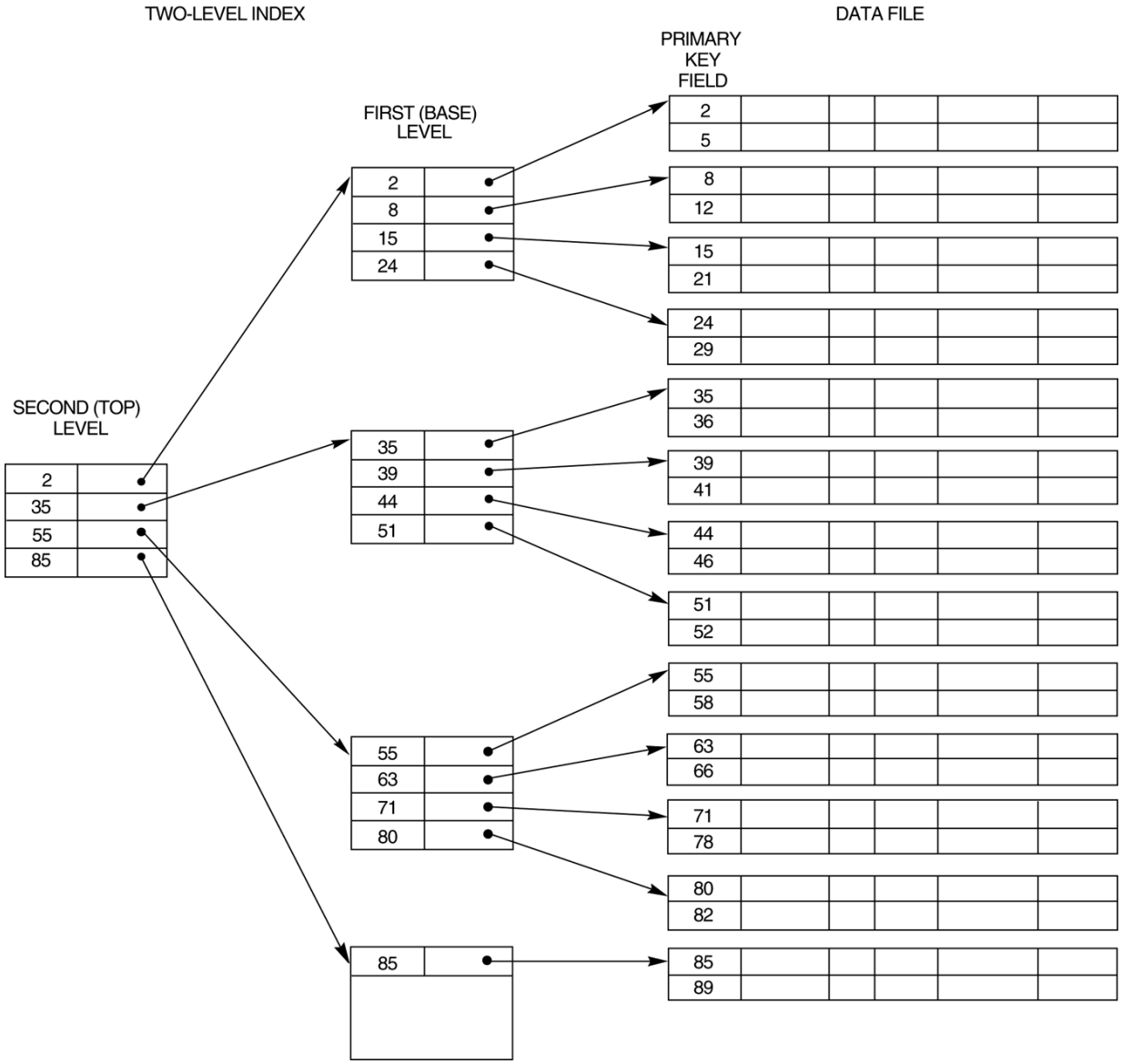
FIGURE 14.6

A two-level primary index resembling ISAM (Indexed Sequential Access Method) organization.



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Multi-Level Indexes

Example: Given the following data file:
EMPLOYEE(NAME, SSN, ADDRESS, JOB, SAL, ...)
Suppose that:
The previous example is converted into a multilevel index

Then, we get:
Index blocking factor Bfr_I= 68 entries/block
number of first-level index blocks b_1= (r/ Bfr_I)= (30000/68)= 442 blocks
number of second-level index blocks b_2= (b_1/ Bfr_I)= (442/68)= 7 blocks
number of third-level index blocks b_3= (b_2/ Bfr_I)= (7/68)= 1 block

multilevel index search needs 1+1+1=3 block accesses, total 3+1=4 block accesses

This is compared to secondary index search cost of:
10 block accesses

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Multi-Level Indexes

- Such a multi-level index is a form of *search tree* ; however, insertion and deletion of new index entries is a severe problem because every level of the index is an *ordered file*.

- Difference from other well-known tree data structures
  - AVL trees, Splay trees, Red-Black trees, AA trees

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FIGURE 14.8

A node in a search tree with pointers to subtrees below it.

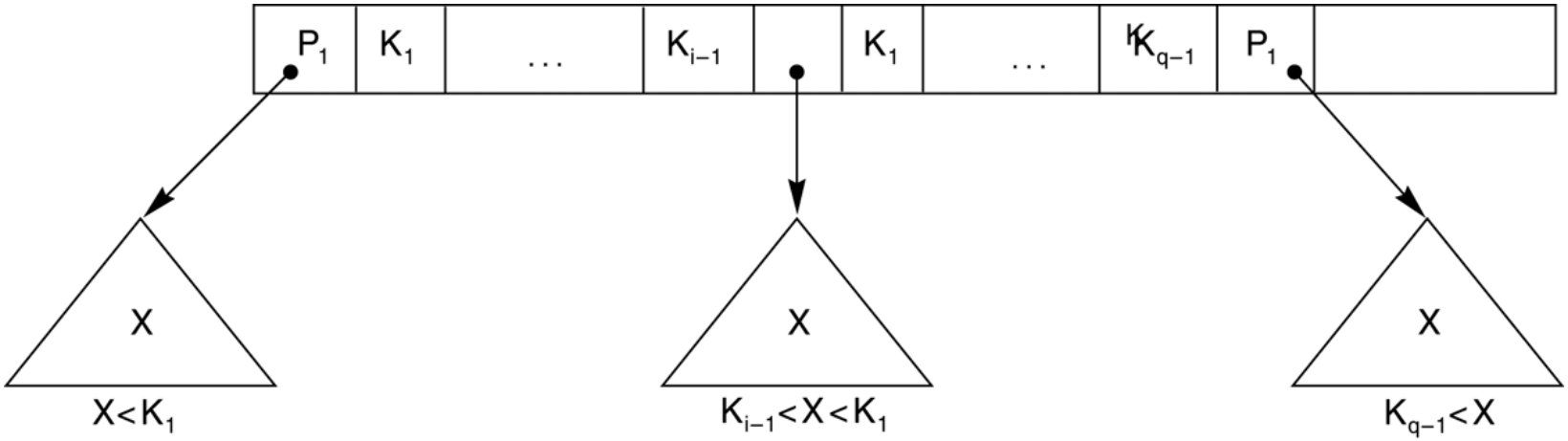

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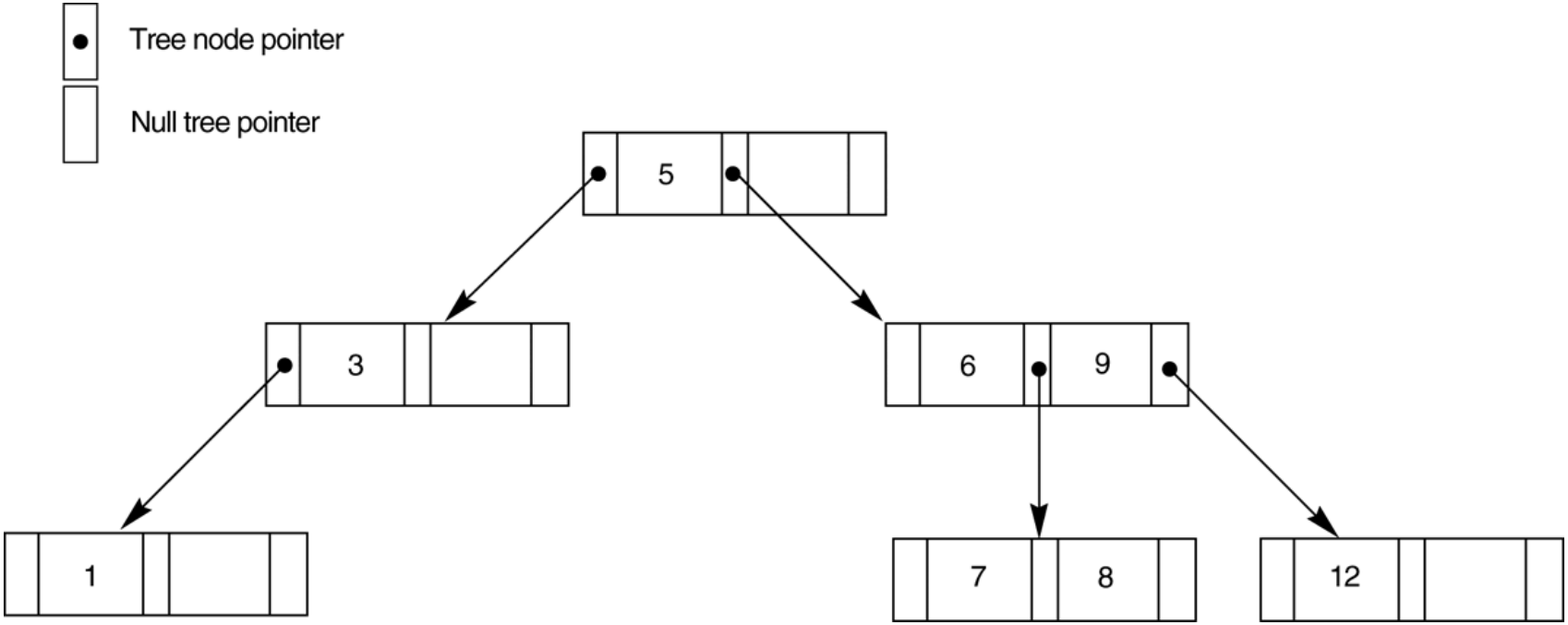
FIGURE 14.9
A search tree of order p = 3.



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Dynamic Multilevel Indexes Using B-Trees and B+-Trees

- Because of the insertion and deletion problem, most multi-level indexes use B-tree or B+-tree data structures, which leave space in each tree node (disk block) to allow for new index entries
- These data structures are variations of search trees that allow efficient insertion and deletion of new search values.
- In B-Tree and B+-Tree data structures, each node corresponds to a disk block
- Each node is kept between half-full and completely full

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Dynamic Multilevel Indexes Using B-Trees and B+-Trees (contd.)

- An insertion into a node that is not full is quite efficient; if a node is full the insertion causes a split into two nodes
- Splitting may propagate to other tree levels
- A deletion is quite efficient if a node does not become less than half full
- If a deletion causes a node to become less than half full, it must be merged with neighboring nodes

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Difference between B-tree and B+-tree

- In a B-tree, pointers to data records exist at all levels of the tree
- In a B+-tree, all pointers to data records exists at the leaf-level nodes
- A B+-tree can have less levels (or higher capacity of search values) than the corresponding B-tree

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FIGURE 14.10

B-tree structures. (a) A node in a B-tree with q – 1 search values. (b) A B-tree of order p = 3. The values were inserted in the order 8, 5, 1, 7, 3, 12, 9, 6.

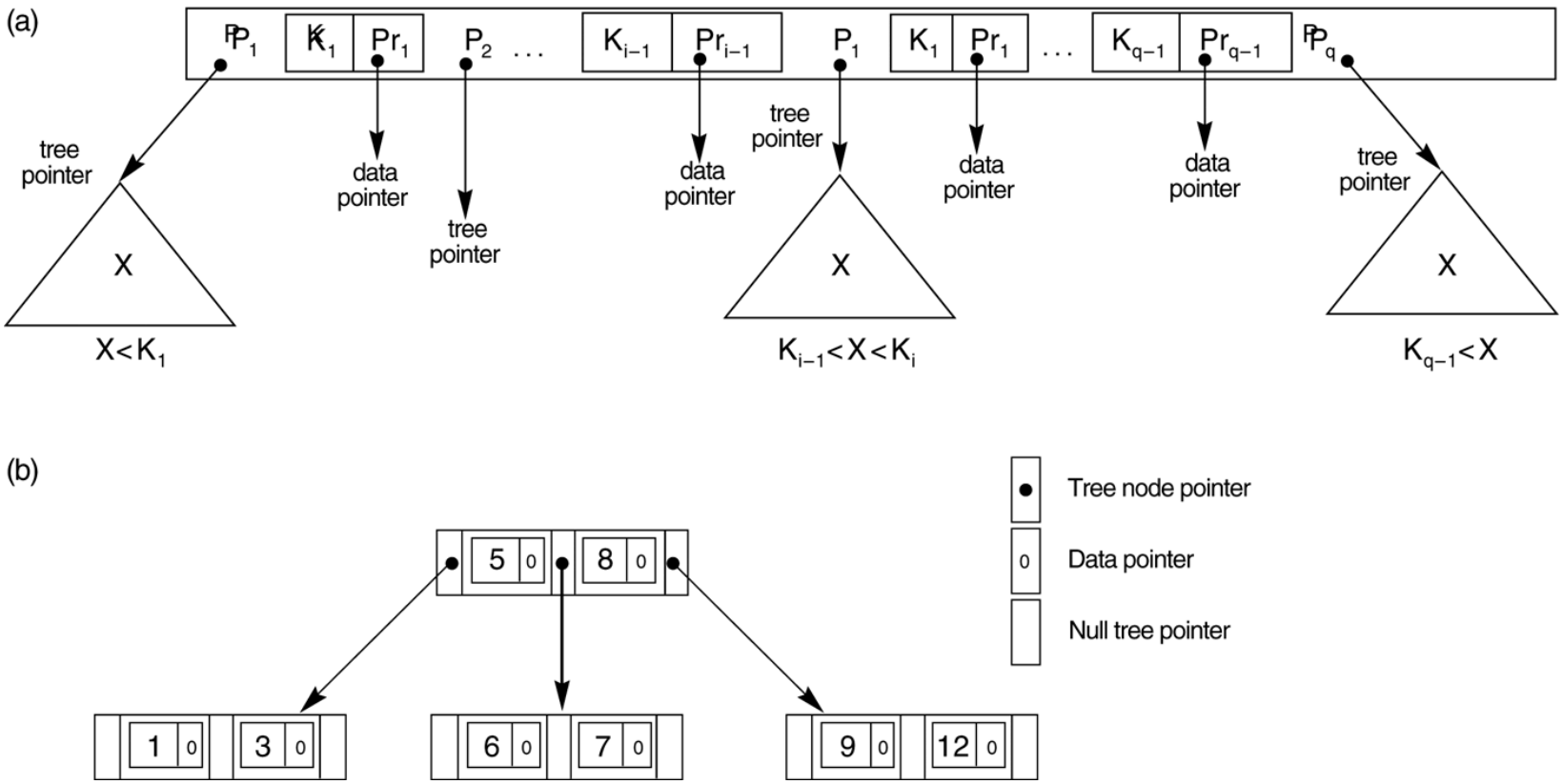

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FIGURE 14.11

The nodes of a B+-tree. (a) Internal node of a B+-tree with q –1 search values. (b) Leaf node of a B+-tree with q – 1 search values and q – 1 data pointers.

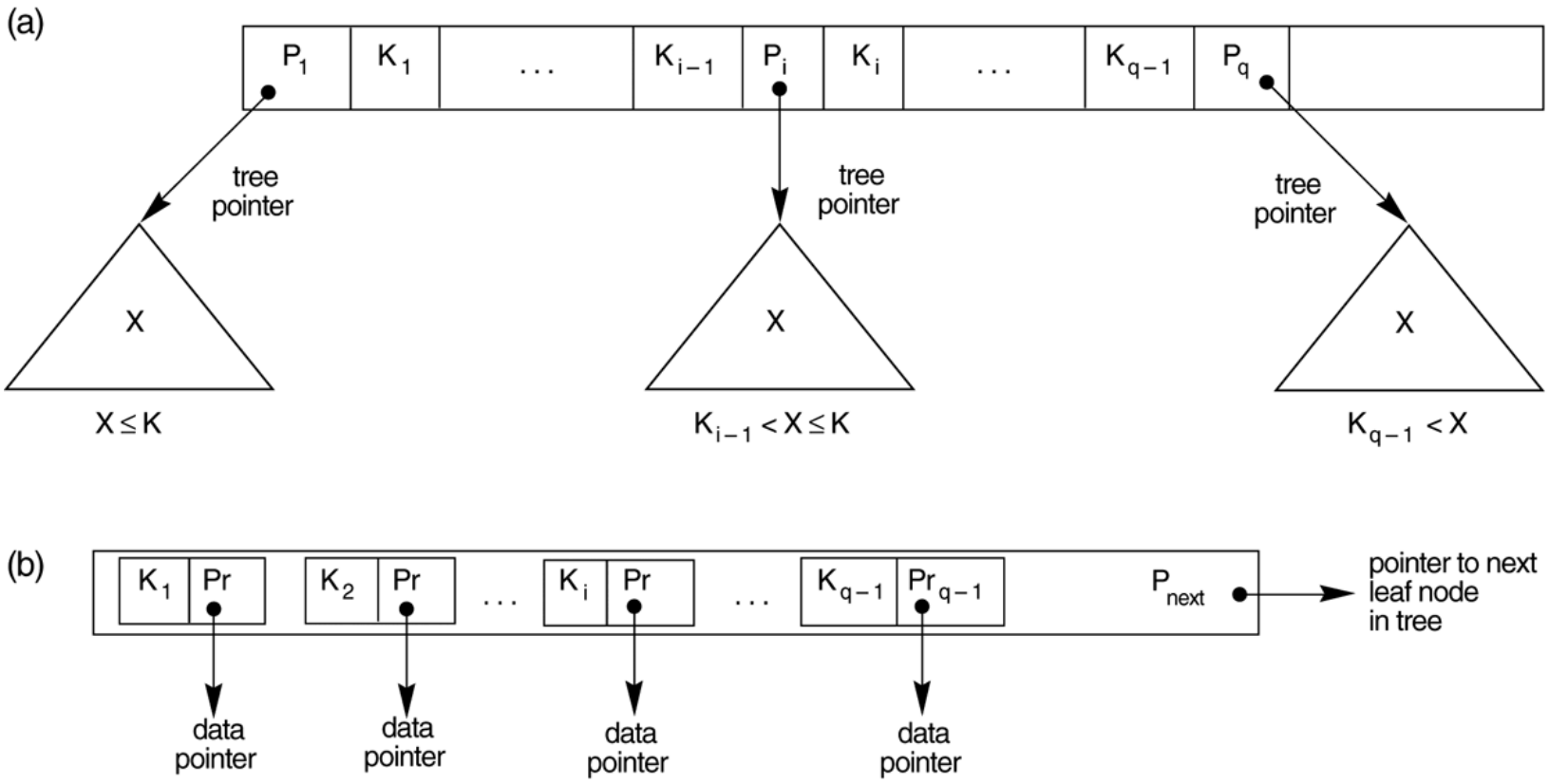

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FIGURE 14.12 **An example of insertion in a B+-tree with q = 3 and p_{leaf} = 2.**

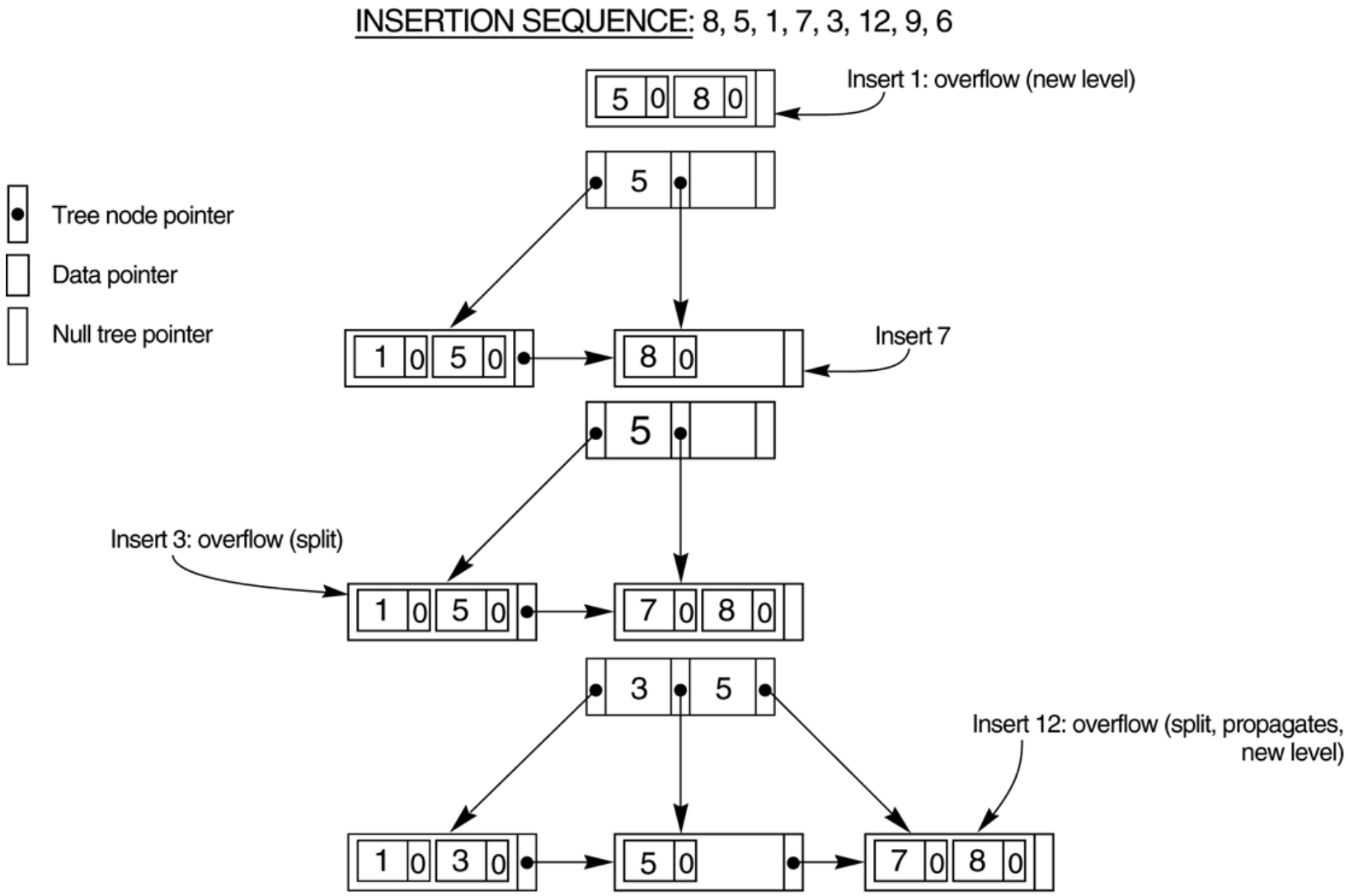

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FIGURE 14.12 **An example of insertion in a B+-tree with q = 3 and p_{leaf} = 2.**

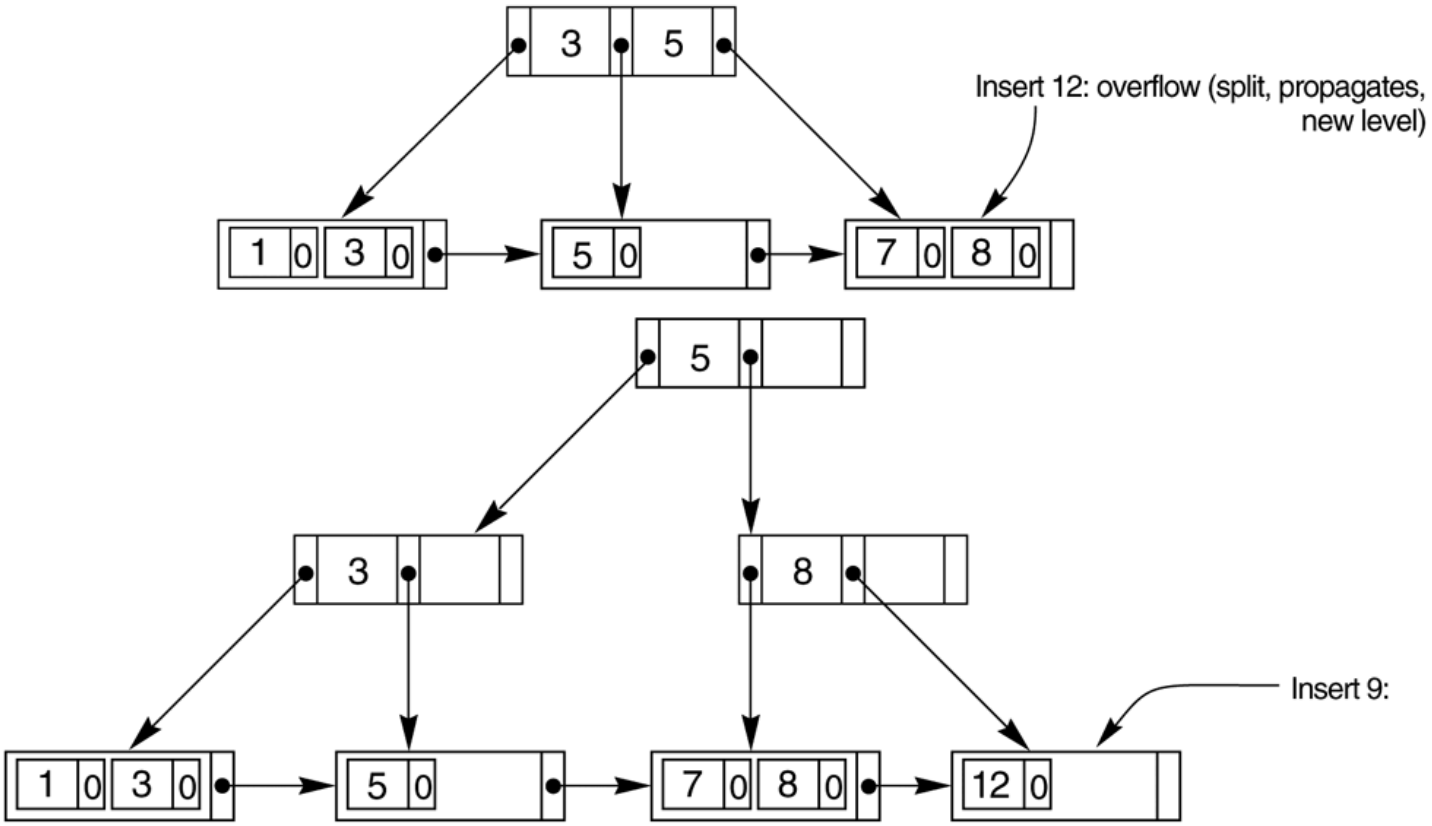

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FIGURE 14.12 **An example of insertion in a B+-tree with q = 3 and p_{leaf} = 2.**

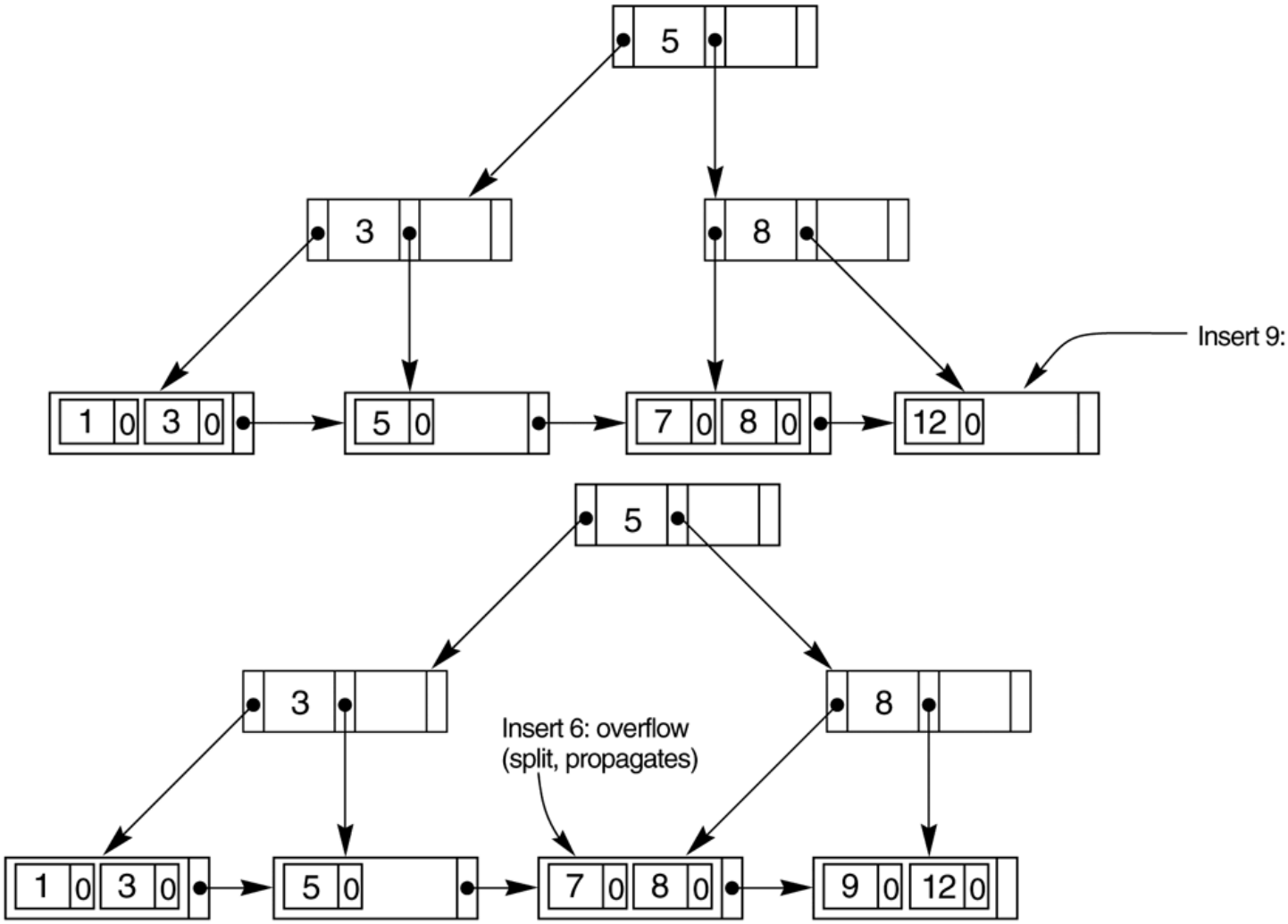

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FIGURE 14.12 **An example of insertion in a B+-tree with q = 3 and p_{leaf} = 2.**

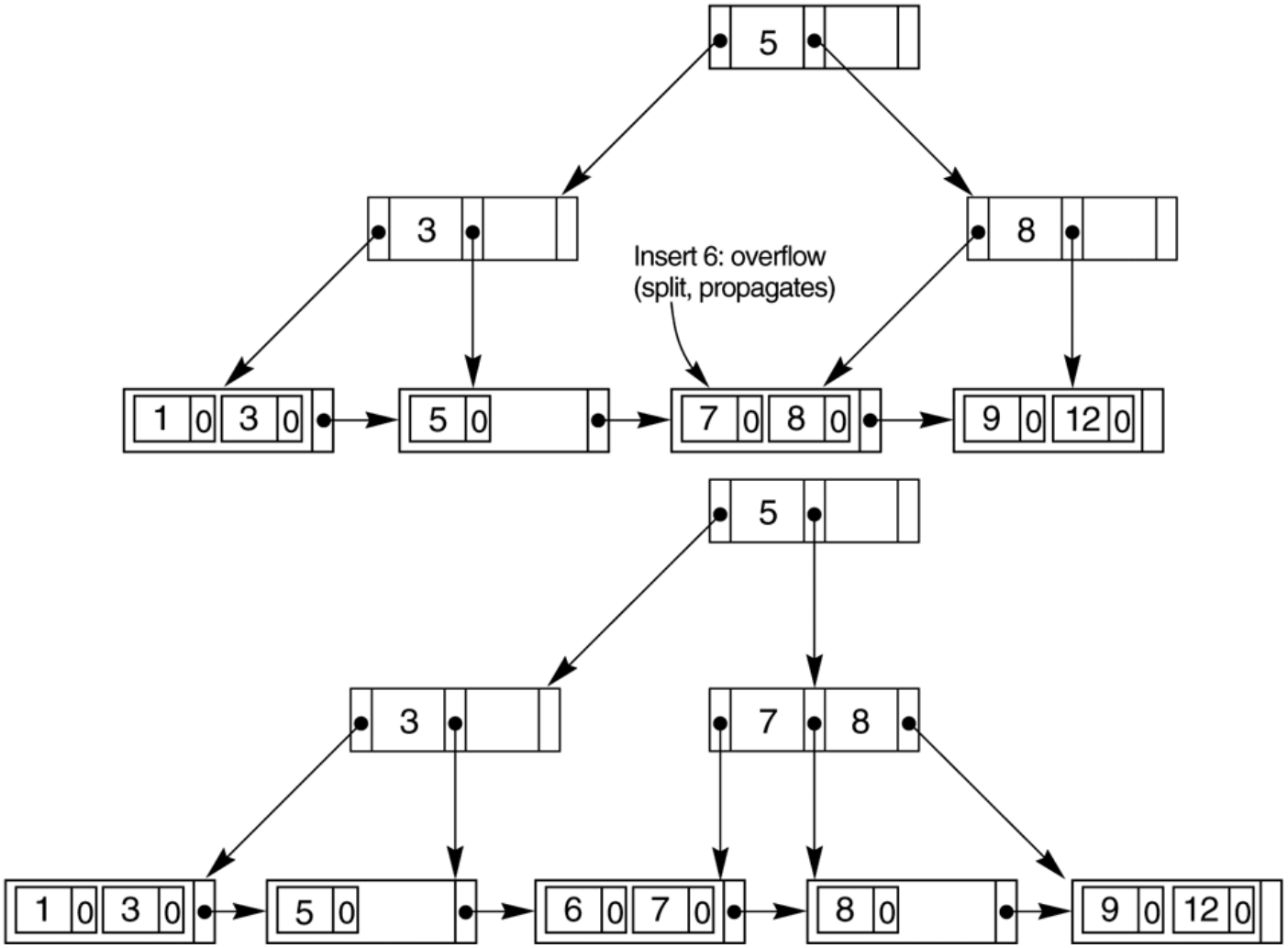

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FIGURE 14.13 **An example of deletion from a B+-tree.**

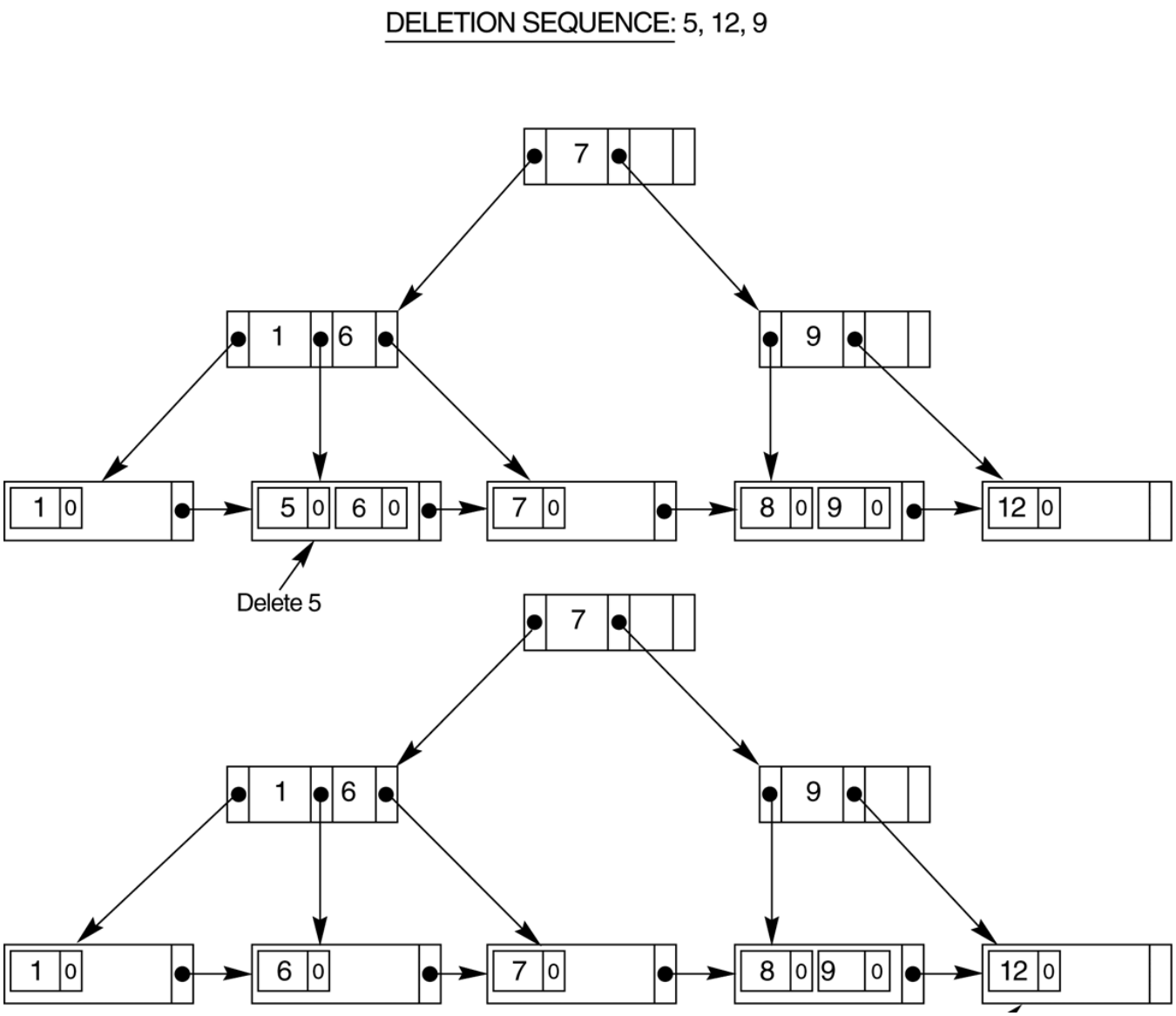

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FIGURE 14.13 **An example of deletion from a B+-tree.**

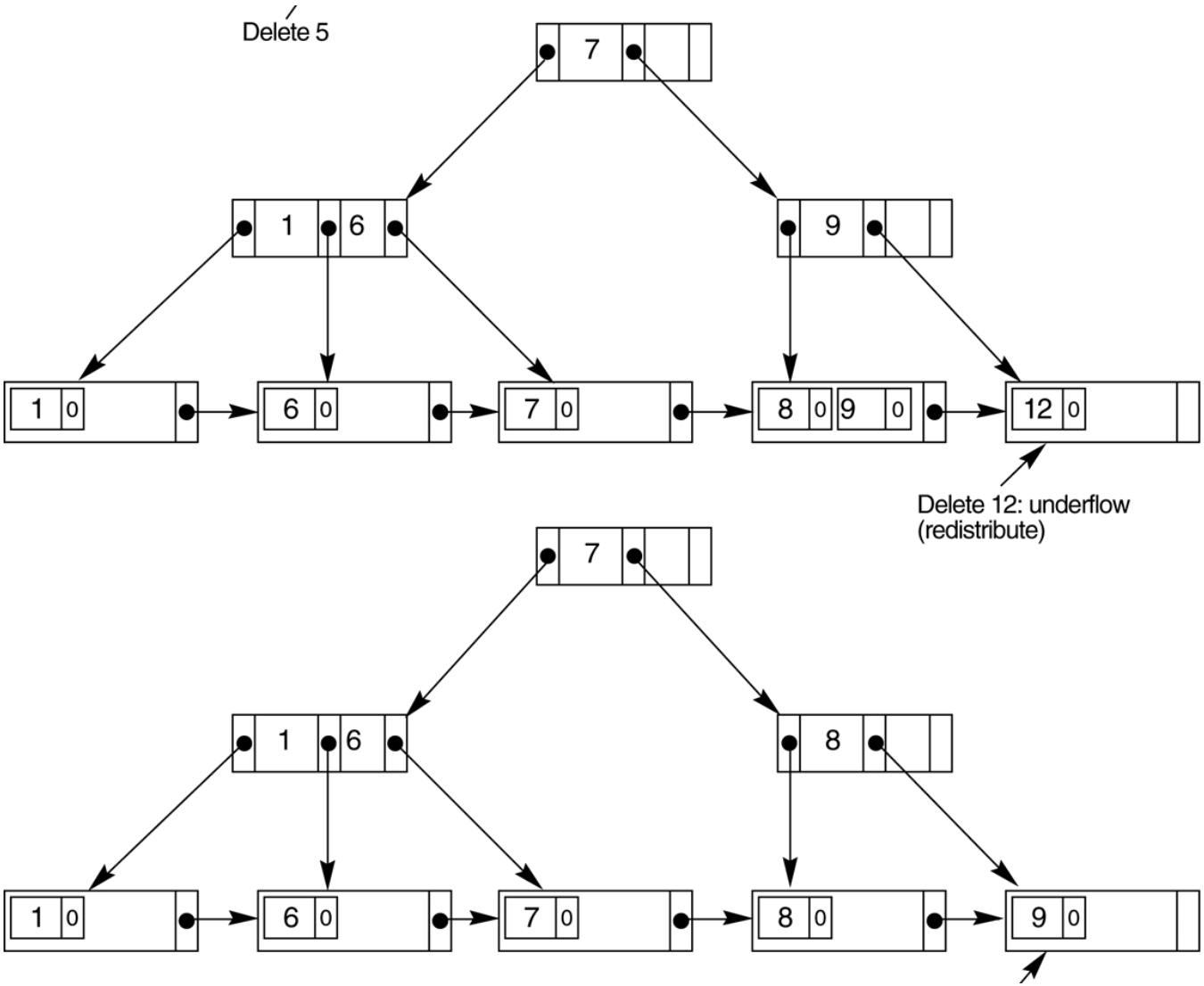

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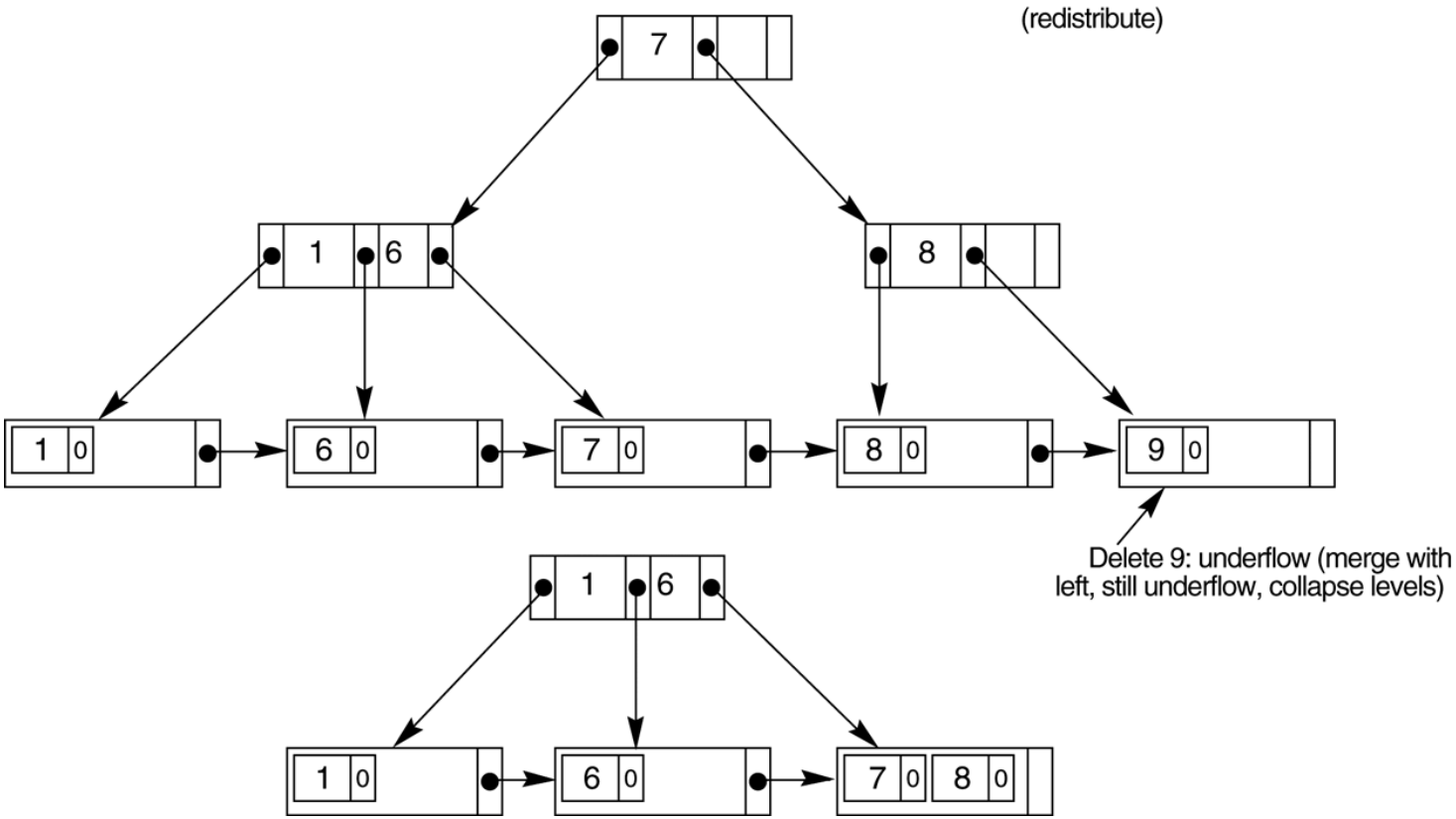
FIGURE 14.13 **An example of deletion from a B+-tree.**



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And, …

- Indexes on Multiple Keys
  - Ordered index on multiple attributes
  - Partitioned hashing
  - Grid Files
- Other types of indexes
  - Using hashing and other data structures as indexes
  - Logical versus physical indexes

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